

(a) Regler:

$$G_R(z) = \frac{T}{T_1} \cdot \frac{z}{z-1}$$

Strecken:

$$G_S(s) = \frac{1-e^{-sT}}{s} \cdot e^{-sT_1} \Rightarrow G_S(z) = (1-z^{-1})z^{-2} \cdot Z\left\{\frac{1}{s}\right\}$$

$$G_S(z) = \frac{z-1}{z} z^{-2} \frac{z}{z-1} = z^{-2}$$

$$G_0(z) = G_R(z) \cdot G_S(z) = \frac{T}{T_1} \cdot \frac{z}{z-1} \cdot z^{-2}$$

$$G_0(z) = \frac{T}{T_1} \cdot \frac{1}{z(z-1)} = \frac{z_0}{M_0}$$

charakteristische Gleichung:

$$N(z) = z_0 + M_0 = \frac{T}{T_1} + z^2 - z = \underbrace{z^2}_{a} - \underbrace{z}_{b} + \underbrace{\frac{T}{T_1}}_c \stackrel{!}{=} 0$$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

aperiodisch, wenn " $\sqrt{0}$ "

$$\Rightarrow b^2 \stackrel{!}{=} 4ac$$

$$\Rightarrow 1 \stackrel{!}{=} 4 \frac{T}{T_1} \Rightarrow \underline{T_1 = 4T}$$

(b)

$$G_R(z) = \frac{\bar{Y}}{E} = \frac{T}{4T} \cdot \frac{z}{z-1} = \frac{1}{4} \cdot \frac{1}{1-z^{-1}} \Rightarrow \bar{Y}(1-z^{-1}) = \frac{1}{4} \bar{E}$$

$$\Rightarrow \underline{y_k = \frac{1}{4} e_k + y_{k-1}}$$